

A hyperbolic approximation of the Cahn-Hilliard equation

Firas Dhaouadi
Università degli Studi di Trento

Joint work with
Sergey Gavriluk (Aix-Marseille University)
Michael Dumbser (University of Trento)



UNIVERSITÀ
DI TRENTO



Finanziato
dall'Unione europea
NextGenerationEU

February 20th, 2025

Cahn-Hilliard equations (1958)

The Cahn-Hilliard equation is postulated as a conservative diffusion equation which writes

$$\frac{\partial c}{\partial t} = \Delta (c^3 - c - \gamma \Delta c) .$$

- $c \in [-1, 1]$ is the order parameter indicating the phases.
- $\gamma \ll 1$ is such that $\sqrt{\gamma}$ is the diffuse interface characteristic length.
- describes well the process of phase separation in binary systems: spinodal decomposition, Ostwald Ripening phenomena, etc
- Has applications for modeling binary alloys, sedimentation problems, etc ...

About the equation

$$\frac{\partial c}{\partial t} = \Delta (c^3 - c - \gamma \Delta c) .$$

Cool features

- scalar PDE.
- Well-posed. [Elliott, Songmu 1986]
- diffuse-interface model (able to deal with strong topological changes).

About the equation

$$\frac{\partial c}{\partial t} = \Delta (c^3 - c - \gamma \Delta c) .$$

Cool features

- scalar PDE.
- Well-posed. [Elliott, Songmu 1986]
- diffuse-interface model (able to deal with strong topological changes).

Not so cool features

- non-convex equation of state (Requires very careful numerical treatment)
- 4th Order in space (explicit solvers: NO)
- Violates principle of Causality (Laplace operator)

Plan of presentation

- 1 On the Cahn-Hilliard equations
- 2 Hyperbolic Model Derivation
 - 2nd-order approximation
 - 1st-order approximation
 - Analysis
- 3 Numerical scheme and Results
 - Numerical scheme
 - Some results

Conservative form and chemical potential

The Cahn-Hilliard equation can be cast into a conservation-law form which writes

$$\frac{\partial c}{\partial t} + \operatorname{div}(\mathbf{j}) = 0, \quad (1)$$

where the mass flux $\mathbf{j}$ is assumed to obey a generalized Fick's law such that

$$\mathbf{j} = -\nabla \mu,$$

and μ is the chemical potential of the system given by

$$\mu = \frac{\delta f}{\delta c} = \frac{\partial f}{\partial c} - \operatorname{div} \left(\frac{\partial f}{\partial \nabla c} \right) = c^3 - c - \gamma \Delta c,$$

where

$$f(c, \nabla c) = \frac{(c^2 - 1)^2}{4} + \frac{\gamma}{2} \|\nabla c\|^2,$$

Lyapunov functional

CH equation admits the Lyapunov functional

$$F(c, \nabla c) = \int_{\mathcal{D}} f(c, \nabla c) \, d\Omega = \int_{\mathcal{D}} \frac{(c^2 - 1)^2}{4} + \frac{\gamma}{2} \|\nabla c\|^2 \, d\Omega$$

Indeed, we have

$$\frac{\partial f}{\partial t} + \operatorname{div}(\mathcal{H}) = -\|\nabla \mu\|^2, \quad \mathcal{H} = \mu \mathbf{j} - \frac{\partial f}{\partial \nabla c} \Delta \mu$$

which in integral form writes

$$\frac{\partial F}{\partial t} = - \int_{\mathcal{D}} \|\nabla \mu\|^2 \, d\Omega \leq 0,$$

Lyapunov functional

CH equation admits the Lyapunov functional

$$F(c, \nabla c) = \int_{\mathcal{D}} f(c, \nabla c) \, d\Omega = \int_{\mathcal{D}} \frac{(c^2 - 1)^2}{4} + \frac{\gamma}{2} \|\nabla c\|^2 \, d\Omega$$

Indeed, we have

$$\frac{\partial f}{\partial t} + \operatorname{div}(\mathcal{H}) = -\|\nabla \mu\|^2, \quad \mathcal{H} = \mu \mathbf{j} - \frac{\partial f}{\partial \nabla c} \Delta \mu$$

which in integral form writes

$$\frac{\partial F}{\partial t} = - \int_{\mathcal{D}} \|\nabla \mu\|^2 \, d\Omega \leq 0,$$

provided suitable boundary conditions.

Hyperbolic reformulation

1 On the Cahn-Hilliard equations

2 Hyperbolic Model Derivation

- 2nd-order approximation
- 1st-order approximation
- Analysis

3 Numerical scheme and Results

- Numerical scheme
- Some results

Modified action functional

Starting from the Lyapunov Functional

$$F(c, \nabla c) = \int_{\mathcal{D}} f(c, \nabla c) \, d\Omega = \int_{\mathcal{D}} \frac{(c^2 - 1)^2}{4} + \frac{\gamma}{2} \|\nabla c\|^2 \, d\mathcal{D},$$

Modified action functional

Starting from the Lyapunov Functional

$$F(c, \nabla c) = \int_{\mathcal{D}} f(c, \nabla c) \, d\Omega = \int_{\mathcal{D}} \frac{(c^2 - 1)^2}{4} + \frac{\gamma}{2} \|\nabla c\|^2 \, d\mathcal{D},$$

let us introduce the following action functional

$$a = \int_t \int_{\mathcal{D}} \mathcal{L} \, d\mathcal{D} dt$$

where

$$\mathcal{L} \left(c, \varphi, \nabla \varphi, \frac{\partial \varphi}{\partial t} \right) = -\frac{(c^2 - 1)^2}{4} - \frac{\gamma}{2} \|\nabla \varphi\|^2 - \frac{\alpha}{2} (c - \varphi)^2 + \frac{\beta}{2} \left(\frac{\partial \varphi}{\partial t} \right)^2.$$

Modified action functional

Starting from the Lyapunov Functional

$$F(c, \nabla c) = \int_{\mathcal{D}} f(c, \nabla c) \, d\Omega = \int_{\mathcal{D}} \frac{(c^2 - 1)^2}{4} + \frac{\gamma}{2} \|\nabla c\|^2 \, d\mathcal{D},$$

let us introduce the following action functional

$$a = \int_t \int_{\mathcal{D}} \mathcal{L} \, d\mathcal{D} dt$$

where

$$\mathcal{L} \left(c, \varphi, \nabla \varphi, \frac{\partial \varphi}{\partial t} \right) = -\frac{(c^2 - 1)^2}{4} - \frac{\gamma}{2} \|\nabla \varphi\|^2 - \frac{\alpha}{2} (c - \varphi)^2 + \frac{\beta}{2} \left(\frac{\partial \varphi}{\partial t} \right)^2.$$

- φ is a new variable substituting c as the order parameter.
- $\alpha \gg 1$ so that $(c - \varphi)$ vanishes in the limit $\alpha \rightarrow +\infty$.
- $\beta \ll 1$ is a small parameter.

Generalized Fick's law for c

$$\mathcal{L}\left(c, \varphi, \nabla \varphi, \frac{\partial \varphi}{\partial t}\right) = -\frac{(c^2 - 1)^2}{4} - \frac{\gamma}{2} \|\nabla \varphi\|^2 - \frac{\alpha}{2} (c - \varphi)^2 + \frac{\beta}{2} \left(\frac{\partial \varphi}{\partial t}\right)^2.$$

Generalized Fick's law now becomes

$$\frac{\partial c}{\partial t} + \operatorname{div}(-\nabla \mu) = 0, \quad \mu = -\frac{\delta \mathcal{L}}{\delta c}$$

Generalized Fick's law for c

$$\mathcal{L}\left(c, \varphi, \nabla \varphi, \frac{\partial \varphi}{\partial t}\right) = -\frac{(c^2 - 1)^2}{4} - \frac{\gamma}{2} \|\nabla \varphi\|^2 - \frac{\alpha}{2}(c - \varphi)^2 + \frac{\beta}{2} \left(\frac{\partial \varphi}{\partial t}\right)^2.$$

Generalized Fick's law now becomes

$$\frac{\partial c}{\partial t} + \operatorname{div}(-\nabla \mu) = 0, \quad \mu = -\frac{\delta \mathcal{L}}{\delta c} = -\frac{\partial \mathcal{L}}{\partial c} = c^3 - c + \alpha(c - \varphi),$$

Generalized Fick's law for c

$$\mathcal{L}\left(c, \varphi, \nabla \varphi, \frac{\partial \varphi}{\partial t}\right) = -\frac{(c^2 - 1)^2}{4} - \frac{\gamma}{2} \|\nabla \varphi\|^2 - \frac{\alpha}{2}(c - \varphi)^2 + \frac{\beta}{2} \left(\frac{\partial \varphi}{\partial t}\right)^2.$$

Generalized Fick's law now becomes

$$\frac{\partial c}{\partial t} + \operatorname{div}(-\nabla \mu) = 0, \quad \mu = -\frac{\delta \mathcal{L}}{\delta c} = -\frac{\partial \mathcal{L}}{\partial c} = c^3 - c + \alpha(c - \varphi),$$

$\Rightarrow$ 2nd-order PDE, no 4th-order terms

$$\frac{\partial c}{\partial t} - \Delta (c^3 - c + \alpha(c - \varphi)) = 0, \quad (I)$$

Euler-Lagrange equation for φ

$$\mathcal{L}\left(\mathbf{c}, \varphi, \nabla \varphi, \frac{\partial \varphi}{\partial t}\right) = -\frac{(c^2 - 1)^2}{4} - \frac{\gamma}{2} \|\nabla \varphi\|^2 - \frac{\alpha}{2} (\mathbf{c} - \varphi)^2 + \frac{\beta}{2} \left(\frac{\partial \varphi}{\partial t}\right)^2.$$

For φ , we simply write the Euler-Lagrange equations.

$$\frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial \varphi_t} \right) + \operatorname{div} \left(\frac{\partial \mathcal{L}}{\partial \nabla \varphi} \right) = \frac{\partial \mathcal{L}}{\partial \varphi}.$$

which gives

$$\beta \frac{\partial^2 \varphi}{\partial t^2} - \operatorname{div} (\gamma \nabla \varphi) = \alpha (c - \varphi) \quad (II)$$

2nd-order approximation of the Cahn-Hilliard equation

Thus so far we have obtained the following system of two 2nd order PDEs

$$\frac{\partial c}{\partial t} - \Delta (c^3 - c + \alpha(c - \varphi)) = 0, \quad (I)$$

$$\beta \frac{\partial^2 \varphi}{\partial t^2} - \gamma \Delta \varphi = \alpha(c - \varphi). \quad (II)$$

2nd-order approximation of the Cahn-Hilliard equation

Thus so far we have obtained the following system of two 2nd order PDEs

$$\frac{\partial c}{\partial t} - \Delta (c^3 - c + \alpha(c - \varphi)) = 0, \quad (I)$$

$$\beta \frac{\partial^2 \varphi}{\partial t^2} - \gamma \Delta \varphi = \alpha(c - \varphi). \quad (II)$$

- Equation (I) is reminiscent of heat equation.
 $\Rightarrow$ Cattaneo-type relaxation.

2nd-order approximation of the Cahn-Hilliard equation

Thus so far we have obtained the following system of two 2nd order PDEs

$$\frac{\partial c}{\partial t} - \Delta (c^3 - c + \alpha(c - \varphi)) = 0, \quad (I)$$

$$\beta \frac{\partial^2 \varphi}{\partial t^2} - \gamma \Delta \varphi = \alpha(c - \varphi). \quad (II)$$

- Equation (I) is reminiscent of heat equation.
 $\Rightarrow$ Cattaneo-type relaxation.
- Equation (II) is a hyperbolic wave equation with right-hand side. $\Rightarrow$ Order reduction.

Order reduction for (II)

$$\beta \frac{\partial^2 \varphi}{\partial t^2} - \operatorname{div}(\gamma \nabla \varphi) = \alpha(c - \varphi) \quad (II)$$

Let us denote the independent variables

$$w = \beta \frac{\partial \varphi}{\partial t}, \quad \mathbf{p} = \nabla \varphi.$$

Order reduction for (II)

$$\beta \frac{\partial^2 \varphi}{\partial t^2} - \operatorname{div}(\gamma \nabla \varphi) = \alpha(c - \varphi) \quad (II)$$

Let us denote the independent variables

$$w = \beta \frac{\partial \varphi}{\partial t}, \quad \mathbf{p} = \nabla \varphi.$$

Therefore (II) becomes

$$\begin{aligned} \frac{\partial w}{\partial t} - \operatorname{div}(\gamma \mathbf{p}) &= -\alpha(\varphi - c), \\ \frac{\partial \varphi}{\partial t} &= \frac{1}{\beta} w, \\ \frac{\partial \mathbf{p}}{\partial t} - \frac{1}{\beta} \nabla w &= 0. \end{aligned}$$

Relaxation for equation (I)

Reminder:

$$\frac{\partial c}{\partial t} - \operatorname{div} (\nabla (c^3 - c + \alpha(c - \varphi))) = 0, \quad (I)$$

which also writes

$$\frac{\partial c}{\partial t} - \operatorname{div} (\nabla \mu) = 0, \quad (I)$$

Relaxation for equation (I)

Reminder:

$$\frac{\partial c}{\partial t} - \operatorname{div} (\nabla (c^3 - c + \alpha(c - \varphi))) = 0, \quad (I)$$

which also writes

$$\frac{\partial c}{\partial t} - \operatorname{div} (\nabla \mu) = 0, \quad (I)$$

$$\begin{aligned} \frac{\partial c}{\partial t} + \operatorname{div} \left(\frac{1}{\tau} \mathbf{q} \right) &= 0, \\ \frac{\partial \mathbf{q}}{\partial t} + \nabla \mu &= -\frac{1}{\tau} \mathbf{q}, \end{aligned}$$

- $\tau \ll 1$ is a relaxation time.
- c is still a conserved quantity in this framework.

Final system approximating the Cahn-Hilliard equations

$$\frac{\partial c}{\partial t} + \operatorname{div} \left(\frac{1}{\tau} \mathbf{q} \right) = 0$$

$$\frac{\partial \mathbf{q}}{\partial t} + \nabla (c^3 - c + \alpha(c - \varphi)) = -\frac{1}{\tau} \mathbf{q}$$

$$\frac{\partial w}{\partial t} - \operatorname{div} (\gamma \mathbf{p}) = -\alpha(\varphi - c)$$

$$\frac{\partial \mathbf{p}}{\partial t} - \frac{1}{\beta} \nabla w = 0$$

$$\frac{\partial \varphi}{\partial t} = \frac{1}{\beta} w$$

- System of hyperbolic equations with relaxations.
- Equations are conservative also in multiple dimensions.

Hyperbolicity

We cast the previous in quasilinear form

$$\frac{\partial \mathbf{Q}}{\partial t} + \mathbf{A}(\mathbf{Q}) \frac{\partial \mathbf{Q}}{\partial x} = \mathbf{S}(\mathbf{Q})$$

where $\mathbf{Q}$ is the vector of conserved variables and $\mathbf{A}(\mathbf{Q})$ is the quasilinear matrix, both given by

$$\mathbf{A}(\mathbf{Q}) = \left(\begin{array}{ccccc|c} 0 & \frac{1}{\tau} & 0 & 0 & 0 & \\ \alpha + 3c^2 - 1 & 0 & 0 & 0 & -\alpha & \\ 0 & 0 & 0 & -\gamma & 0 & \\ 0 & 0 & -\frac{1}{\beta} & 0 & 0 & \\ \hline & \mathbf{O}_{5,5} & & & & \mathbf{O}_{5,4} \end{array} \right),$$

$$\mathbf{Q} = (c, q_1, w, p_1, \varphi, q_2, q_3, p_2, p_3)^T,$$

Eigenvalues

System admits a full set of real eigenvalues given by

$$\chi_1 = -\frac{\sqrt{3c^2 + \alpha - 1}}{\sqrt{\tau}},$$

$$\chi_2 = -\frac{\sqrt{\gamma}}{\sqrt{\beta}},$$

$$\chi_{3-7} = 0,$$

$$\chi_8 = \frac{\sqrt{\gamma}}{\sqrt{\beta}},$$

$$\chi_9 = \frac{\sqrt{3c^2 + \alpha - 1}}{\sqrt{\tau}}.$$

and a corresponding set of linearly independent eigenfields.

Lyapunov Functional

Proposition

The proposed hyperbolic Cahn-Hilliard system admits the following Lyapunov functional

$$E = \int_{\mathcal{D}} e(c, \varphi, \mathbf{q}, \mathbf{p}, w) \, d\Omega,$$
$$e(c, \varphi, \mathbf{q}, \mathbf{p}, w) = \frac{(c^2 - 1)^2}{4} + \frac{\gamma}{2} \|\mathbf{p}\|^2 + \frac{\alpha}{2} (c - \varphi)^2 + \frac{1}{2\beta} w^2 + \frac{1}{2\tau} \|\mathbf{q}\|^2$$

Proof

We express the fluxes as a function of the conjugate variables

$$\frac{\partial c}{\partial t} + \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{q}} \right) = 0$$

$$\frac{\partial \mathbf{q}}{\partial t} + \nabla \left(\frac{\partial e}{\partial c} \right) = - \frac{\partial e}{\partial \mathbf{q}}$$

$$\frac{\partial w}{\partial t} - \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{p}} \right) = - \frac{\partial e}{\partial \varphi}$$

$$\frac{\partial \mathbf{p}}{\partial t} - \nabla \left(\frac{\partial e}{\partial w} \right) = 0$$

$$\frac{\partial \varphi}{\partial t} = \frac{\partial e}{\partial w}$$

Proof

We express the fluxes as a function of the conjugate variables

$$\frac{\partial e}{\partial c} \cdot \left\{ \frac{\partial c}{\partial t} + \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{q}} \right) = 0 \right.$$

$$\frac{\partial e}{\partial \mathbf{q}} \cdot \left\{ \frac{\partial \mathbf{q}}{\partial t} + \nabla \left(\frac{\partial e}{\partial c} \right) = - \frac{\partial e}{\partial \mathbf{q}} \right.$$

$$\frac{\partial e}{\partial w} \cdot \left\{ \frac{\partial w}{\partial t} - \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{p}} \right) = - \frac{\partial e}{\partial \varphi} \right.$$

$$\frac{\partial e}{\partial \mathbf{p}} \cdot \left\{ \frac{\partial \mathbf{p}}{\partial t} - \nabla \left(\frac{\partial e}{\partial w} \right) = 0 \right.$$

$$\frac{\partial e}{\partial \varphi} \cdot \left\{ \frac{\partial \varphi}{\partial t} = \frac{\partial e}{\partial w} \right.$$

Proof

We express the fluxes as a function of the conjugate variables

$$\frac{\partial e}{\partial c} \cdot \left\{ \frac{\partial c}{\partial t} + \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{q}} \right) = 0 \right.$$

$$\frac{\partial e}{\partial \mathbf{q}} \cdot \left\{ \frac{\partial \mathbf{q}}{\partial t} + \nabla \left(\frac{\partial e}{\partial c} \right) = - \frac{\partial e}{\partial \mathbf{q}} \right.$$

$$\frac{\partial e}{\partial w} \cdot \left\{ \frac{\partial w}{\partial t} - \operatorname{div} \left(\frac{\partial e}{\partial \mathbf{p}} \right) = - \frac{\partial e}{\partial \varphi} \right.$$

$$\frac{\partial e}{\partial \mathbf{p}} \cdot \left\{ \frac{\partial \mathbf{p}}{\partial t} - \nabla \left(\frac{\partial e}{\partial w} \right) = 0 \right.$$

$$\frac{\partial e}{\partial \varphi} \cdot \left\{ \frac{\partial \varphi}{\partial t} = \frac{\partial e}{\partial w} \right.$$

$$\Rightarrow \frac{\partial e}{\partial t} + \operatorname{div} \left(\frac{\partial e}{\partial c} \frac{\partial e}{\partial \mathbf{q}} - \frac{\partial e}{\partial \mathbf{p}} \frac{\partial e}{\partial w} \right) = - \left\| \frac{\partial e}{\partial \mathbf{q}} \right\|^2 \leq 0,$$

Energy conservation: exchange form

$$e(c, \varphi, \mathbf{q}, \mathbf{p}, w) = e_I(c, \varphi, \mathbf{q}) + e_{II}(\mathbf{p}, w),$$

$$\begin{cases} e_I = \frac{(c^2 - 1)^2}{4} + \frac{\alpha}{2}(c - \varphi)^2 + \frac{1}{2\tau} \|\mathbf{q}\|^2, \\ e_{II} = \frac{\gamma}{2} \|\mathbf{p}\|^2 + \frac{1}{2\beta} w^2, \end{cases}$$

one can obtain the following evolution equation for each of the energy parts

$$\frac{\partial e_I}{\partial t} + \operatorname{div} \left(\frac{\partial e_I}{\partial c} \frac{\partial e_I}{\partial \mathbf{q}} \right) = - \frac{\partial e_I}{\partial \varphi} \frac{\partial e_{II}}{\partial w} - \left\| \frac{\partial e_I}{\partial \mathbf{q}} \right\|^2,$$

$$\frac{\partial e_{II}}{\partial t} - \operatorname{div} \left(\frac{\partial e_{II}}{\partial \mathbf{p}} \frac{\partial e_{II}}{\partial w} \right) = \frac{\partial e_I}{\partial \varphi} \frac{\partial e_{II}}{\partial w},$$

Numerical methods

In order to solve the model numerically and also compare it with reference solutions, we propose here:

- ① A numerical scheme for the original Cahn-Hilliard equation based on 4th order implicit conservative finite differences
- ② Explicit MUSCL-Hancock for the hyperbolic approximation.

Implicit conservative finite differences for CH

We propose here a semi-implicit conservative in order to solve numerically the original Cahn-Hilliard equations. We rewrite the latter as follows

$$\frac{\partial c}{\partial t} - \operatorname{div}(\mathbf{F}) + \gamma \Delta^2 c = 0$$

where $\mathbf{F}$ is the flux given by

$$\mathbf{F} = \chi(c) \nabla c, \quad \chi(c) = 3c^2 - 1$$

The scheme writes

$$c_{i,j}^{n+1} = c_{i,j}^n + \frac{\Delta t}{\Delta x} \left(\mathcal{F}_{i+\frac{1}{2},j}^{n+1} - \mathcal{F}_{i-\frac{1}{2},j}^{n+1} \right) + \frac{\Delta t}{\Delta y} \left(\mathcal{G}_{i,j+\frac{1}{2}}^{n+1} - \mathcal{G}_{i,j-\frac{1}{2}}^{n+1} \right) - \gamma \Delta \Delta_h c_{i,j}^{n+1}.$$

Computation of the intercell fluxes

The intercell fluxes $\mathcal{F}_{i+\frac{1}{2},j}^{n+1}$ and $\mathcal{G}_{i,j+\frac{1}{2}}^{n+1}$, in the x and y directions respectively, are computed using finite-differences as follows

$$\mathcal{F}_{i+\frac{1}{2},j}^{n+1} = \chi_{i+\frac{1}{2},j}^{n+1,r} (\nabla_x c)_{i+\frac{1}{2},j}^{n+1},$$

$$\begin{cases} \chi_{i+\frac{1}{2},j}^{n+1,r} \simeq \frac{1}{12} \left(7 \chi_{i,j}^{n+1,r} - \chi_{i-1,j}^{n+1,r} + 7 \chi_{i+1,j}^{n+1,r} - \chi_{i+2,j}^{n+1,r} \right) \\ (\nabla_x c)_{i+\frac{1}{2},j}^{n+1} \simeq -\frac{1}{12 \Delta x} \left(15 c_{i-1,j}^{n+1} - 15 c_{i,j}^{n+1} + c_{i+1,j}^{n+1} - c_{i+2,j}^{n+1} \right) \end{cases}$$

(similarly for $\mathcal{G}_{i,j+\frac{1}{2}}^{n+1}$)

Computation of the intercell fluxes

The intercell fluxes $\mathcal{F}_{i+\frac{1}{2},j}^{n+1}$ and $\mathcal{G}_{i,j+\frac{1}{2}}^{n+1}$, in the x and y directions respectively, are computed using finite-differences as follows

$$\mathcal{F}_{i+\frac{1}{2},j}^{n+1} = \chi_{i+\frac{1}{2},j}^{n+1,r} (\nabla_x c)_{i+\frac{1}{2},j}^{n+1},$$

$$\begin{cases} \chi_{i+\frac{1}{2},j}^{n+1,r} \simeq \frac{1}{12} \left(7 \chi_{i,j}^{n+1,r} - \chi_{i-1,j}^{n+1,r} + 7 \chi_{i+1,j}^{n+1,r} - \chi_{i+2,j}^{n+1,r} \right) \\ (\nabla_x c)_{i+\frac{1}{2},j}^{n+1} \simeq -\frac{1}{12 \Delta x} \left(15 c_{i-1,j}^{n+1} - 15 c_{i,j}^{n+1} + c_{i+1,j}^{n+1} - c_{i+2,j}^{n+1} \right) \end{cases}$$

(similarly for $\mathcal{G}_{i,j+\frac{1}{2}}^{n+1}$)

These are 4th order approximations.

Discretization of the bi-laplacian operator

$\Delta\Delta_h c_{i,j}^{n+1}$ is a discretization of the bi-Laplacian operator in the cell-centers as follows

$$\begin{aligned}\Delta\Delta_h c_{i,j}^{n+1} = & -\frac{\Delta t}{\Delta x^4} \left(c_{i-2,j}^{n+1} - 4c_{i-1,j}^{n+1} + 6c_{i,j}^{n+1} - 4c_{i+1,j}^{n+1} + c_{i+2,j}^{n+1} \right) \\ & -\frac{\Delta t}{\Delta y^4} \left(c_{i,j-2}^{n+1} - 4c_{i,j-1}^{n+1} + 6c_{i,j}^{n+1} - 4c_{i,j+1}^{n+1} + c_{i,j+2}^{n+1} \right) \\ & -\frac{2\Delta t}{\Delta x^2 \Delta y^2} \left(c_{i-1,j-1}^{n+1} - 2c_{i,j-1}^{n+1} + c_{i+1,j-1}^{n+1} - 2c_{i-1,j}^{n+1} \right. \\ & \quad \left. + 4c_{i,j}^{n+1} - 2c_{i+1,j}^{n+1} + c_{i-1,j+1}^{n+1} - 2c_{i,j+1}^{n+1} + c_{i+1,j+1}^{n+1} \right)\end{aligned}$$

Discretization of the bi-laplacian operator

$\Delta\Delta_h c_{i,j}^{n+1}$ is a discretization of the bi-Laplacian operator in the cell-centers as follows

$$\begin{aligned}\Delta\Delta_h c_{i,j}^{n+1} = & -\frac{\Delta t}{\Delta x^4} \left(c_{i-2,j}^{n+1} - 4c_{i-1,j}^{n+1} + 6c_{i,j}^{n+1} - 4c_{i+1,j}^{n+1} + c_{i+2,j}^{n+1} \right) \\ & -\frac{\Delta t}{\Delta y^4} \left(c_{i,j-2}^{n+1} - 4c_{i,j-1}^{n+1} + 6c_{i,j}^{n+1} - 4c_{i,j+1}^{n+1} + c_{i,j+2}^{n+1} \right) \\ & -\frac{2\Delta t}{\Delta x^2 \Delta y^2} \left(c_{i-1,j-1}^{n+1} - 2c_{i,j-1}^{n+1} + c_{i+1,j-1}^{n+1} - 2c_{i-1,j}^{n+1} \right. \\ & \quad \left. + 4c_{i,j}^{n+1} - 2c_{i+1,j}^{n+1} + c_{i-1,j+1}^{n+1} - 2c_{i,j+1}^{n+1} + c_{i+1,j+1}^{n+1} \right)\end{aligned}$$

System over all the domain is then solved using GMRES (Matrix is not symmetric positive-definite ...)

Brief overview of the numerical method for hyperbolic approximation

- Explicit second-order MUSCL-Hancock scheme.
- We used FORCE or Rusanov approximate Riemann solvers

$$\mathbf{Q}_{i,j}^{n+1} = \mathbf{Q}_{i,j}^n - \frac{\Delta t}{\Delta x} \left(\mathcal{F}_{i+\frac{1}{2},j} - \mathcal{F}_{i-\frac{1}{2},j} \right) - \frac{\Delta t}{\Delta y} \left(\mathcal{G}_{i,j+\frac{1}{2}} - \mathcal{G}_{i,j-\frac{1}{2}} \right) + \Delta t \mathbf{S} \left(\mathbf{Q}_{i,j}^n + \frac{\Delta t}{2} \partial_t \mathbf{Q}_{i,j} \right).$$

Intercelle fluxes are computed by using

$$\mathcal{F}_{i+\frac{1}{2},j} = \frac{1}{2} \left(\mathbf{F}(\tilde{\mathbf{W}}_{i,j}^R) + \mathbf{F}(\tilde{\mathbf{W}}_{i,j}^L) \right) - \frac{1}{2} \lambda_{max}^{(x)} \left(\tilde{\mathbf{W}}_{i,j}^R - \tilde{\mathbf{W}}_{i,j}^L \right)$$

Where $\tilde{\mathbf{W}}_{i,j}^R$ are piecewise linear reconstructions of the solution in the cell (i, j) .

Summary of numerical methods

Original



a semi-implicit
conservative FD scheme on
staggered grids with 4th
order FD for Laplace operators

Hyperbolic



Explicit
Muscl-Hancock

Convergence in α : ODE for original system

CH equation in 1D:

$$\frac{\partial c}{\partial t} + \frac{\partial J}{\partial x} = 0, \quad J = \gamma \frac{\partial^3 c}{\partial x^3} - (3c^2 - 1) \frac{\partial c}{\partial x}$$

Convergence in α : ODE for original system

CH equation in 1D:

$$\frac{\partial c}{\partial t} + \frac{\partial J}{\partial x} = 0, \quad J = \gamma \frac{\partial^3 c}{\partial x^3} - (3c^2 - 1) \frac{\partial c}{\partial x}$$

Stationary states:

$$\begin{aligned} \frac{\partial J}{\partial x} &= 0, \quad J = \gamma \frac{\partial^3 c}{\partial x^3} - (3c^2 - 1) \frac{\partial c}{\partial x}, \\ J(x_0) &= J^0, \quad c(x_0) = c^0, \quad \left. \frac{\partial c}{\partial x} \right|_{x=x_0} = c_I^0, \quad \left. \frac{\partial^2 c}{\partial x^2} \right|_{x=x_0} = c_{II}^0, \end{aligned}$$

which can be written in a first-order system as

Convergence in α : ODE for original system

CH equation in 1D:

$$\frac{\partial c}{\partial t} + \frac{\partial J}{\partial x} = 0, \quad J = \gamma \frac{\partial^3 c}{\partial x^3} - (3c^2 - 1) \frac{\partial c}{\partial x}$$

Stationary states:

$$\begin{aligned} \frac{\partial J}{\partial x} &= 0, \quad J = \gamma \frac{\partial^3 c}{\partial x^3} - (3c^2 - 1) \frac{\partial c}{\partial x}, \\ J(x_0) &= J^0, \quad c(x_0) = c^0, \quad \left. \frac{\partial c}{\partial x} \right|_{x=x_0} = c_I^0, \quad \left. \frac{\partial^2 c}{\partial x^2} \right|_{x=x_0} = c_{II}^0, \end{aligned}$$

which can be written in a first-order system as

$$\begin{aligned} c' &= c_I, \quad c_I' = c_{II}, \quad c_{II}' = \frac{1}{\gamma} (J + (3c^2 - 1)c_1), \quad J' = 0, \\ c(x_0) &= c^0, \quad c_I(x_0) = c_I^0, \quad c_{II}(x_0) = c_{II}^0, \quad J(x_0) = J^0, \end{aligned}$$

Hyperbolic counterpart

$$\begin{aligned} \varphi' &= p, \quad p' = \frac{\alpha}{\gamma} (\varphi - c), \quad c' = \frac{\alpha p - q/\tau}{3c^2 - 1 + \alpha}, \quad q' = 0, \\ \varphi(x_0) &= c^0 + \frac{\gamma}{\alpha} c_{II}^0, \quad p(x_0) = c_I^0, \quad c(x_0) = c^0, \quad q(x_0) = -\tau J^0, \end{aligned}$$

Reminder about original system ODE

$$\begin{aligned} c' &= c_I, \quad c_I' = c_{II}, \quad c_{II}' = \frac{1}{\gamma} (J + (3c^2 - 1)c_1), \quad J' = 0, \\ c(x_0) &= c^0, \quad c_I(x_0) = c_I^0, \quad c_{II}(x_0) = c_{II}^0, \quad J(x_0) = J^0, \end{aligned}$$

Convergence in α

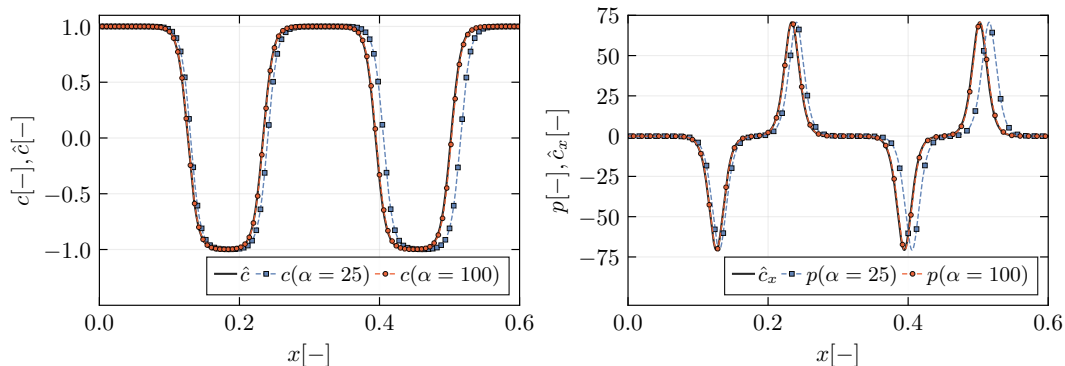


Figure 1: Comparison of a stationary solution of the hyperbolic Cahn-Hilliard model (discontinuous lines) with the original counterpart (solid line) for different values of the penalty parameter α .

Convergence table in α

α	$\ c - \hat{c}\ _{L_2}$	$\ \mathbf{p} - \nabla \hat{c}\ _{L_2}$	$\ c - \varphi\ _{L_2}$	$\mathcal{O}(c - \hat{c})$	$\mathcal{O}(\mathbf{p} - \nabla \hat{c})$	$\mathcal{O}(c - \varphi)$
25	2.64×10^{-1}	5.66×10^{-1}	7.01×10^{-3}	—	—	—
50	1.35×10^{-1}	3.02×10^{-1}	3.51×10^{-3}	0.96	0.90	0.99
100	6.82×10^{-2}	1.54×10^{-1}	1.75×10^{-3}	0.99	0.97	0.99
400	1.70×10^{-2}	3.86×10^{-2}	4.39×10^{-4}	1.00	0.99	0.99
1600	3.80×10^{-3}	8.64×10^{-3}	1.10×10^{-4}	1.08	1.08	1.00

Table 1: Convergence table for the L_2 errors when comparing the numerical Cauchy problem solutions for the hyperbolic model with the the original Cahn-Hilliard equation.

Exact solution for the original equation

One can find a family of exact one-dimensional stationary periodic solutions to the Cahn-Hilliard system expressed as

$$c_{\epsilon}(x) = \sqrt{1-\epsilon} \operatorname{sn} \left(\sqrt{\frac{\epsilon+1}{2\gamma}}(x-x_0), \sqrt{\frac{1-\epsilon}{1+\epsilon}} \right)$$

Here, $\operatorname{sn}(x, s)$ is the Jacobi elliptic sine function, and s is the elliptic modulus.
 $\epsilon \in [0, 1]$.

It is worthy of note that in the limit $\epsilon \rightarrow 0$ corresponding to $s \rightarrow 1$, one recovers the well-known solution

$$c(x) = \tanh \left(\frac{x-x_0}{\sqrt{2\gamma}} \right)$$

as a particular case.

Exact elliptic function solution

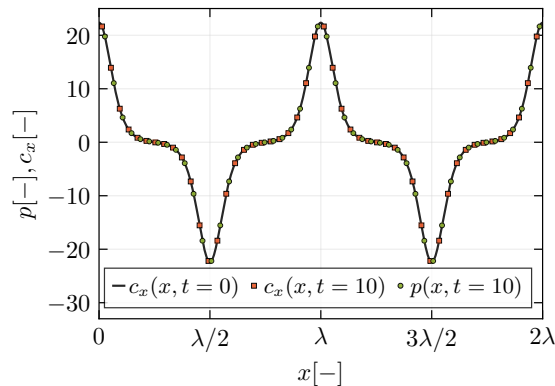
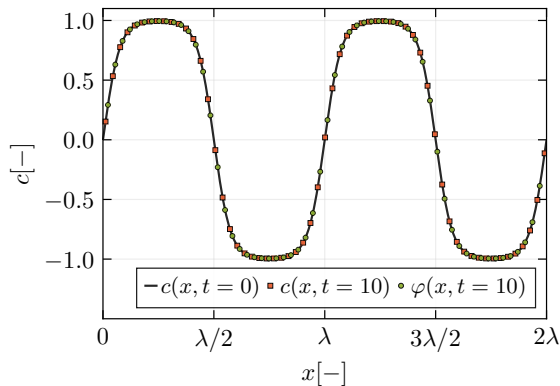


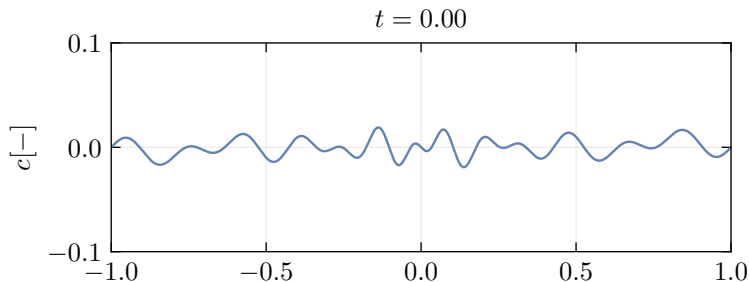
Figure 2: $\gamma = 0.001$. Computational domain is $[0, 2\lambda]$, discretized over $N = 2000$ cells. $\beta = 10^{-6}$, $\alpha = 500$ and $\tau = 8 \cdot 10^{-4}$. CFL = 0.95 and final simulation time is $t = 10$.

Spinodal decomposition

We suggest the following initial data

$$c(x) = \begin{cases} 0.01 \left((\sin(10\pi(1+x))) - \sin(10\pi(1+x)^2) \right), & \text{if } x \in [-1, 0] \\ -0.01 \left((\sin(10\pi(1-x))) - \sin(10\pi(1-x)^2) \right), & \text{if } x \in [0, 1]. \end{cases}$$

This function is built in such a way that it is C^∞ over $[-1, 1]$ as well as over $\mathbb{R}$ by periodic prolongation.



Spinodal decomposition ($\gamma = 10^{-3}, \beta = 10^{-7}, \alpha = 500, \tau = 10^{-5}$)

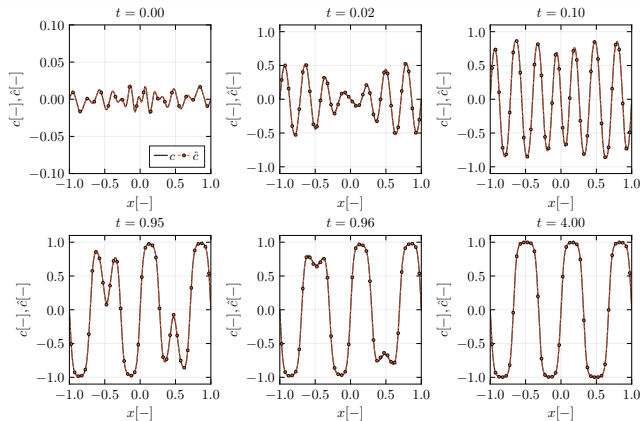
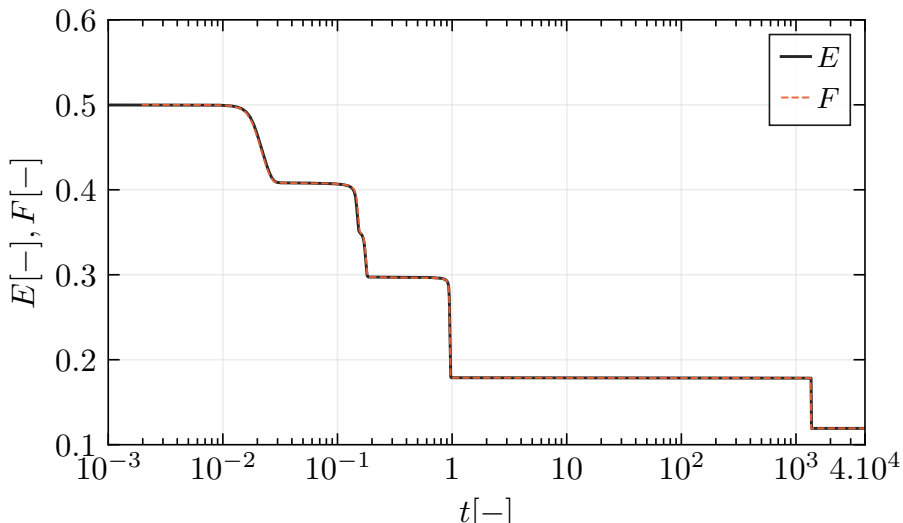


Figure 3: Comparison of the numerical results between the original model (orange) and its hyperbolic counterpart (black). $N = 2000$ computational cells.

Long-time behavior of the Lyapunov functionals



Long-time behavior of the Lyapunov functionals

Lyapunov functional of the Cahn-Hilliard equation

$$F(c, \nabla c) = \int_{\mathcal{D}} \frac{(c^2 - 1)^2}{4} + \frac{\gamma}{2} \|\nabla c\|^2 \, d\Omega,$$

Lyapunov functional of the hyperbolic version

$$E = \int_{\mathcal{D}} \frac{(c^2 - 1)^2}{4} + \frac{\gamma}{2} \|\mathbf{p}\|^2 + \frac{\alpha}{2} (c - \varphi)^2 + \frac{1}{2\beta} w^2 + \frac{1}{2\tau} \|\mathbf{q}\|^2 \, d\Omega,$$

Ostwald Ripening in 1D ($\gamma = 10^{-3}, \beta = 10^{-7}, \alpha = 500, \tau = 10^{-4}$)

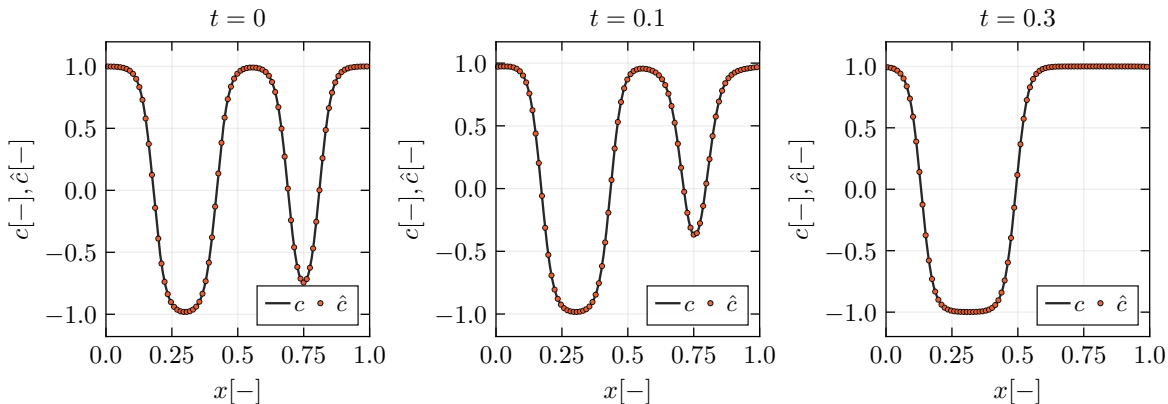


Figure 5: Comparison of the numerical solutions for hyperbolic Cahn-Hilliard model (black line) and the original model (red dots) for the Ostwald Ripening test case at times $t = \{0, 0.1, 0.3\}$.

Stationary Bubble

we numerically solve the time-dependent Cahn-Hilliard equation in radial coordinates

$$\frac{\partial c}{\partial t} - \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \left(c^3 - c - \frac{\gamma}{r} \frac{\partial}{\partial r} \left(r \frac{\partial c}{\partial r} \right) \right) \right) = 0, \quad r = \sqrt{x^2 + y^2},$$

and we consider the following initial guess

$$c_0(r) = -\tanh \left(\frac{r - 0.5}{\sqrt{2\gamma}} \right),$$

We use the same scheme presented earlier for the original Cahn-Hilliard equations.

Stationary Bubble

we numerically solve the time-dependent Cahn-Hilliard equation in radial coordinates

$$\frac{\partial c}{\partial t} - \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \left(c^3 - c - \frac{\gamma}{r} \frac{\partial}{\partial r} \left(r \frac{\partial c}{\partial r} \right) \right) \right) = 0, \quad r = \sqrt{x^2 + y^2},$$

and we consider the following initial guess

$$c_0(r) = -\tanh \left(\frac{r - 0.5}{\sqrt{2\gamma}} \right),$$

We use the same scheme presented earlier for the original Cahn-Hilliard equations.
(Pseudo-transient continuation methods).

Stationary bubble($\gamma = 10^{-3}, \beta = 10^{-7}, \alpha = 1000, \tau = 10^{-5}$)

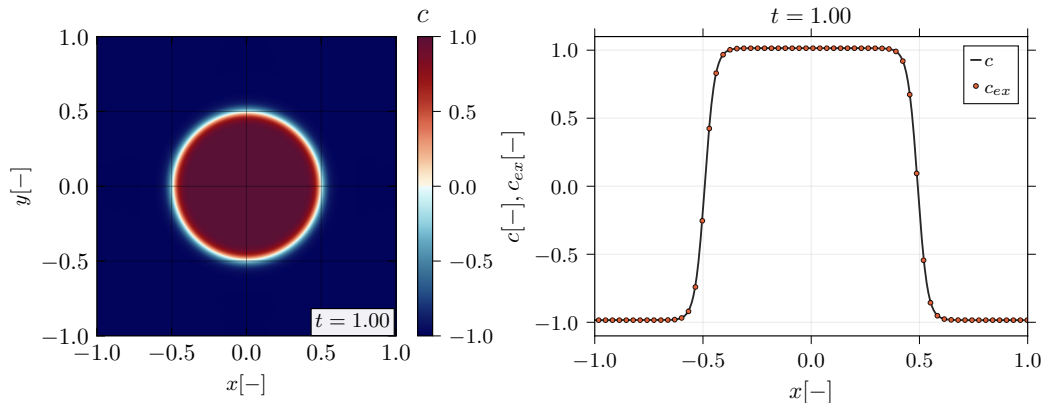
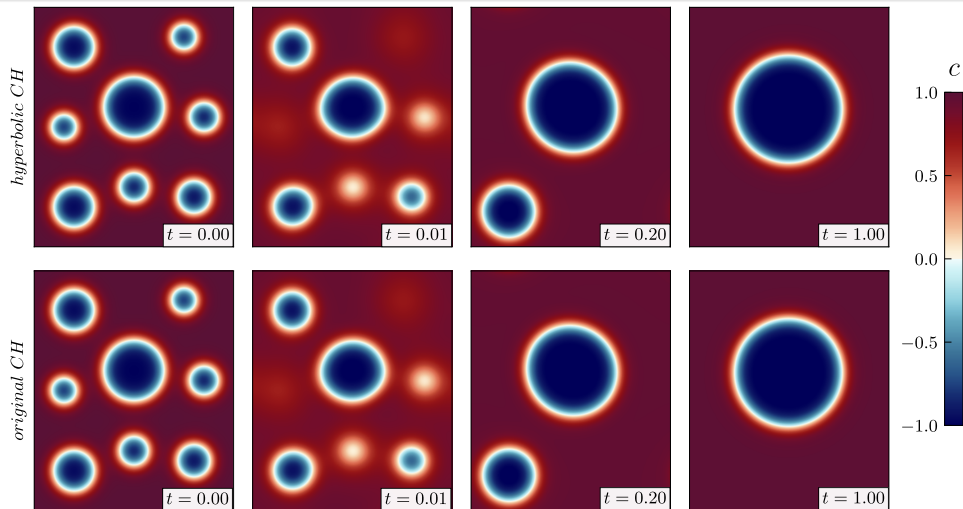


Figure 6: Left: 2D color plot of the radial stationary solution. Right: radial cut of the numerical solution along the line $y = 0$ at $t = 1$ (black line), compared with the exact solution c_{ex} , provided as initial data (red dots). Domain is 500×500

Ostwald Ripening in 2D($\gamma = 10^{-3}, \beta = 10^{-7}, \alpha = 500, \tau = 10^{-5}$)



Ostwald Ripening in 2D : horizontal Cuts

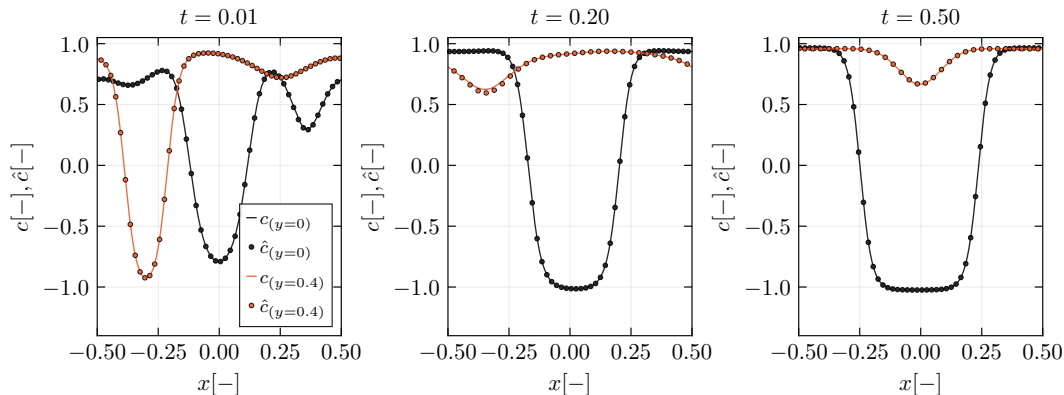


Figure 7: Horizontal cuts over the lines $y = 0$ (red) and $y = 0.4$ (black). Domain is 600×720

Conclusion and Perspective

- We presented a new formulation for an approximate hyperbolic Cahn-Hilliard system.
- An original scheme was conceived to solve the original equation using conservative finite differences.
- Comparison of results showed excellent agreement between the results in one and two dimensions.

Perspectives

- Better formulation fully from variational principles if possible.
- Extension to PDEs of the same family (Allen-Cahn, CH-NS, *etc.*)
- Investigation of bound-preservation properties.
- Semi-implicit discretization, asymptotic preserving schemes, time-step optimization, structure-preserving schemes, etc ...

Thank you for your attention !

[1] Dhaouadi F, Dumbser M, Gavriluk S. 2025 **A first-order hyperbolic reformulation of the Cahn-Hilliard equation**. *Proceedings of the Royal Society A* (To appear soon), 20240606. (10.1098/rspa.2024.0606)

The codes and the parameter files for all the test cases are also published as supplementary material:

[2] Dhaouadi, Firas; Dumbser, Michael; Gavriluk, Sergey (2025). Supplementary material from "A first-order hyperbolic reformulation of the Cahn-Hilliard equation". The Royal Society. Collection. <https://doi.org/10.6084/m9.figshare.c.7637355.v1>

Acknowledgement: *This project is supported and funded by NextGeneration EU, Azione 247 MUR Young Researchers – MSCA/SoE.*

Appendix 1: Lyapunov functional and boundary conditions

CH equation admits the Lyapunov functional

$$F(c, \nabla c) = \int_{\mathcal{D}} f(c, \nabla c) \, d\Omega = \int_{\mathcal{D}} \frac{(c^2 - 1)^2}{4} + \frac{\gamma}{2} \|\nabla c\|^2 \, d\Omega$$

Indeed, we have

$$\frac{\partial f}{\partial t} + \operatorname{div}(\mathcal{H}) = -\|\nabla \mu\|^2, \quad \mathcal{H} = \mu \mathbf{j} - \frac{\partial f}{\partial \nabla c} \Delta \mu$$

which in integral form writes

$$\frac{\partial F}{\partial t} = - \int_{\mathcal{D}} \|\nabla \mu\|^2 \, d\Omega \leq 0,$$

Appendix 1: Boundary conditions for Cahn-Hilliard

$$\int_{\partial\mathcal{D}} \mathcal{H} \cdot \mathbf{n} \, d\Sigma = 0 \quad \forall t \geq 0. \quad (6)$$

For instance, this is the case for

$$\mathbf{j} \cdot \mathbf{n} = 0, \quad \text{on } \partial\mathcal{D}, \quad (7a)$$

$$\frac{\partial f}{\partial \nabla c} \cdot \mathbf{n} = \nabla c \cdot \mathbf{n} = 0, \quad \text{on } \partial\mathcal{D}. \quad (7b)$$

In particular, (7a) also conserves the total mass of the system. Note that also periodic boundary conditions yield mass conservation and decay of energy, since fluxes on opposite sides of the domain cancel each other out.